National University of Computer and Emerging Sciences
Lahore Campus

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Technical & Business Writing (SS2007)

Sessional-II Exam

Total Time (Hrs): 1
Total Marks: 35
Total Questions: 1

Date: 02 November, 2026

Course Instructor(s)

Muhammad Zulfiqar Ali

[redacted] Roll No

7ABC Sections

[redacted] Student Signature

Attempt all the questions in the answer copy.

***CLO # 3: Transform** technical writings/reports in IEEE format.*

Q1: A.

i. The following Literature Review section of a research article is missing citations at six points. **Construct and rewrite** its revised version by inserting IEEE citations at the most proper positions. You must first **correlate** the citations with the text as a few extra ones are also given.

ii. **Generate** a suitable title for this research article. **[1.5*6=9 +1= 10 marks]**

Note: Do not change the original text or do any paraphrasing. Only rewrite with correct citations.

Frequency control of power systems is an emerging area of research due to the increasing penetration of variable Renewable Energy Sources (RES) such as wind and solar Photovoltaic (PV) generation [2], [7]. For example, as the inertia decreases with the displacement of conventional synchronous generators there is a concern that frequency excursions become faster, and therefore the likelihood of instability occurring earlier increases [8]. While there are significant ongoing efforts from both academia and industry on how to best deal with the frequency control challenges (during both normal and abnormal system conditions), it is not clear how different real-world RES-dominated power systems compare in terms of the 'strength' of frequency control. This paper attempts to fill this gap by comparing the strength of frequency control of four real-world synchronous islands power systems namely the Great Britain (GB), All-Island Power System (AIPS) of Ireland, and Australia (AUS) mainland, [5], and Tasmania (TAS). The frequency control strength of each system is evaluated by means of various metrics of frequency quality. It is generally understood that small (islanded) synchronous power systems exhibit rapid and substantial frequency excursions compared to large power systems following typical generation loss [6], [3].

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References to select from:

[1]	B. Badrzadeh et al., *Power System Dynamic Modelling and Analysis in Evolving Networks*. Cham, Switzerland: Springer, 2024.
[2]	M. Martínez-Barbeito, D. Gomila, and P. Colet, "Dynamical model for power grid frequency fluctuations: Application to islands with high penetration of wind generation," *IEEE Trans. Sustain. Energy*, vol. 14, no. 3, pp. 1436–1445, Jul. 2023.
[3]	H. Bevrani, H. Golpîra, A. R. Messina, N. Hatziargyriou, F. Milano, and T. Ise, "Power system frequency control: An updated review of current solutions and new challenges," *Electr. Power Syst. Res.*, vol. 194, May 2021, Art. no. 107114.
[4]	M. Nedd et al., "Operating a zero carbon GB power system in 2025: Frequency and fault current [annexes-review of system and network issues, frequency stability, power electronic devices and fault current, & market needs]," Dept. Electron. Elect. Eng., Univ. Strathclyde, Glasgow, U.K., Tech. Rep. 74793, 2020, doi: 10.17868/74793.
[5]	AEMO. (2024). *Frequency Monitoring—Quarter 1 2024*. [Online]. Available: https://aemo.com.au/
[6]	D. Al Kez, A. Foley, F. Ahmed, and J. Morrow, "Overview of frequency control techniques in power systems with high inverter based resources: Challenges and mitigation measures," *IET Smart Grid*, vol. 6, no. 5, pp. 447–469, 2023.
[7]	T. Kërçi, M. Hurtado, S. Tweed, M. V. Escudero, E. Kennedy, and F. Milano, "Emerging challenges of integrating solar PV in the Ireland and northern Ireland power systems," in *Proc. IEEE Power Energy Soc. Gen. Meeting (PESGM)*, Ireland, Jul. 2024, pp. 1–5.
[8]	N. Hatziargyriou et al., "Definition and classification of power system stability—Revisited & extended," *IEEE Trans. Power Syst.*, vol. 36, no. 4, pp. 3271–3281, Jul. 2021.

B. You are working in OGDC Equipment Design Lab. **Develop** for internal management an unsolicited proposal for a new piece of oil-drilling safety equipment. **[20 marks]**

1. **Generate** an Abstract: Only write three bullet points covering Need, Solution, and Benefit. [6]
2. The Body:
   - **a.** Enlist three technical specs.
   - **b.** Three management milestones.
   - **c.** Three items cost budget table in Pk Rs. [9]
3. **Organize** conclusion in a cohesive, persuasive paragraph to ensure fast management approval. [5]

C. Document the following data into a published IEEE journal citation format: **[5 marks]**

N. Zhang,	p. 1842,
Vol. 16, no. 4,	*Energies*,
Y. Huang,	Z. Xu,
"The definition of power grid strength and its calculation methods for power systems with high proportion nonsynchronous-machine sources,"	Feb. 2023.

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